Marmara Üniversitesi
Electrical and Electronics Engineering Department
Spring 2023
EE4091 – Photonics: Research Paper
Due June 10, 2023

Prepare a research paper addressing one of the topics below.

Instructions

- Individual submission (3-4 pages).
- Choose a topic from the list below. Your research should answer the topic question. Focus on the technical details related to the course subjects.
- Reserve your selected topic by sending email (deadline May 28, 2023).
- Paper should consist of an introduction and main topic description with supporting pictures, curves, diagrams, equations, references, etc. Copy-pasted reports will not be credited.

List of topics

1) What is the state-of-the-art efficiency for solid-state lighting?
2) How is the "green gap" covered and by what material systems?
3) Is the future bright for organic LEDs?
4) How are dielectric mirrors designed for maximum reflection?
5) What are the types of LED packaging?
6) What is the technical background of water desalination using UV?
7) How are IR radiations used for high temperature measurements?
8) What are the types and details of lasers used in LASIK surgery for vision correction.
9) Is free-space optical communications feasible?
10) Is zinc-oxide a good semiconductor for photonics?